Yocelyn T. Gutiérrez Guerrero

Ph. D

Curriculum Vitae
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Education

2024-Current **Genomicist - Bioinformatician Research**
Barcelona Institute for Global Health (IS-Global),
Barcelona, Spain
GenMoz Project: Building malaria genomics capacities for women pregnant and children in Mozambique
PI: Alfredo Mayor

2024-Current **Associate-Visiting Research, University of California Berkeley**
Nachman Lab,
Museum of Vertebrate Zoology, Department of Integrative Biology, Berkeley, CA, US
Project: Genomic lanscape and the role of transposon element polymorphism and structural variation in wild house mouse populations across diverse environments
PI: Michael W. Nachman

2023 - 2024 **Postdoctoral Research. University of California, Berkeley**
Nachman Lab,
Museum of Vertebrate Zoology, Department of Integrative Biology, Berkeley, CA, US
Project: Genomic lanscape and the role of transposon element polymorphism and structural variation in wild house mouse populations across diverse environments
PI: Michael W. Nachman

2021 - 2023 **Postdoctoral Research. University of California, Berkeley**
Nachman Lab,
Museum of Vertebrate Zoology, Department of Integrative Biology, Berkeley, CA, US
Project: Genomic basis of environmental adaptation in house mice across North and South hemispheres in America
PI: Michael W. Nachman

2021 **Ph.D. in Biomedical Sciences, Universidad Nacional Autónoma de México (UNAM)**
Biological Sciences Graduate Program, UNAM, Mexico City, Mexico
Project: Genomic basis of dietary diversification and nectar-feeding specialization in Phyllosotmid bats

Publications

Gutiérrez-Guerrero YT., Phifer-Rixey M., Nachman MW. (2024) Across two continents: The genomic basis of environmental adaptation in house mice (*Mus musculus domesticus*) from the Americas. PLoS genetics 20 (7),e1011036 Doi:10.1371/journal.pgen.1011036

Barrera CA., Ortega J., **Gutiérrez-Guerrero YT**., Baeza JA. (2023) Comparative mitochondrial genomics of American nectar-feeding long-nosed bats *Leptonycteris* spp. with insights into the phylogeny of the family Phyllostomidae. Gene 147588 Doi:10.1016/j.gene.2023.147588

Basaldúa S., Rodríguez-Pico M., **Gutiérrez-Guerrero YT**., Ortega J., Baeza JA. (2023) The mitochondrial genome of the endemic and endangered trumpet-nosed bat *Musonycteris harrisoni* (Chiroptera: Phyllostomidae). Journal of Natural History 57:1206-1219
Doi:10.1080/00222933.2022.2150581

Vargas-Trejo KJ., Ortega J., **Gutiérrez-Guerrero YT**., Gutiérrez EG., Baeza A. (2023) The mitochondrial genomes of big eared bats, *Macrotus waterhousii* and *Macrotus californicus* (Chiroptera: Phyllostomidae: Macrotinae). Gene 863:147295

Doi:10.1016/j.gene.2023.147295

Brun A., **Gutiérrez-Guerrero YT**., Caviedes E., Karasov W., Martínez del Río C. (2022) Opportunities lost? The evolution of the absence of trehalose digestion in birds. Physiological and Biochemical Zoology, 95:4
Doi:10.1086/720232

McWhorther TJ., Rader JA., Schondube JE., Fleming PA., **Gutiérrez-Guerrero YT**., Martínez del Río C. (2021) Sucrose digestion capacity in birds shows convergent coevolution with nectar composition across continents. iScience 24:102717
Doi:10.106/j.isci.2021.102717

Moreno-Santillán D., Lama MT., **Gutiérrez-Guerrero YT**., Brown A, Donat P., Zhao H., Rossiter JS., Yohe LR., Potter JH., Teeling EC., Vernes SC., Davies KT., Myers E., Hughes GM., Huang Z., Hoffman F., Corthals AP., Ray D., Dávalos LM. (2021) Large-scale genome sampling reveals unique immunity and metabolic adaptations in bats. Molecular Ecology 00:1-19
Doi:10.111/mec.16027

Barrera-Redondo J., Sánchez-de la Vega G., Aguirre-Liguori JA., Castellanos-Morales G., **Gutiérrez-Guerrero YT**., Aguirre-Dugua X., Aguirre-Planter E., Tenaillon MI., Lira-Saade R., Eguiarte LE. (2021) The domestication of *Cucurbita argyrosperma* as revealed by the genome of its wild relative. Horticultor Research 8(109)
Doi:10.1038/s41438-021-00544-9

Gasca-Pineda J., **Gutiérrez-Guerrero YT**., Aguirre-Planter E., Eguiarte E. (2020) The role of environmental, local adaptation and past climate fluctuations on the amount and distribution of genetic diversity in the Mexican teosintes. American Journal of Botany 107(11): 1542-1554
Doi:10.1002/ajb2.1561

Martínez del Río C & **Gutiérrez-Guerrero YT**. (2020) A Evolutionary remedy for an abominable physiological mystery: Benign hyperglycemia in birds. Journal of Molecular Evolution 88:715-719
Doi:10.1007/s00239-020-09970-0

Gutiérrez-Guerrero YT., Ibarra-Laclette EI., Martínez del Río C., Barrera-Redondo J, Rebollar EA, Ortega J, León-Paniagua L, Urrutia A, Aguirre-Planter E., Eguiarte LE. (2020) Genomic consequences of dietary diversification and parallel evolution due to nectarivory in leaf-nosed bats. GigaScience 9(6)
Doi:10.1093/gigascience

Barrera-Redondo J, Ibarra-Laclette E, Vázquez-Lobo A, **Gutiérrez-Guerrero YT**, et al., (2019) The genome of *Cucurbita argyrosperma* (Silver-seed gourd) reveals faster rates of protein-coding and long noncoding RNA turnover and neofunctionalization within Cucurbita. Molecular Plant 12(4):506-520
Doi:10.1016/j.molp.2018.12.023

Castellanos-Morales G., **Gutiérrez-Guerrero YT**., Gámez N., Eguiarte LE. (2016) Use of molecular and environmental analysis for integrated in situ and ex situ conservation: The case of the Mexican prairie dog. Biological Conservation, 204:284-29
Doi:10.1016/j.biocon.2016.10.036

Publication: Book Chapters

Castellanos-Morales G., **Gutiérrez-Guerrero YT**. (2025) Phylogenomics for the Study of Speciation and Diversification in Neotropical Mammals. In: Melletti, M., Gallina Tessaro, S., Ortega, J., Tirira, D.G. (eds) Mammals of Middle and South America: History, Biogeography, Conservation. Handbook of the Mammals of Middle and South America. Springer, Cham
Doi:10.1007/978-3-031-43163-0_24-1

Gutiérrez-Guerrero YT., Castellanos-Morales G., Eguiarte LE. (2020) Combining mitochondrial and nuclear data for pedigree reconstruction in absence of genealogical information: analysis of a captive population of the endemic and endangered Mexican prairie dog. 1: 371-378. In: Ortega J & Maldonado J (2020) Conservation Genetics in Mammals. Springer, Smithsonian, DC Washington
Doi:10.1007/978-3-030-33334-8_16

Honor thesis

Gutiérrez-Guerrero YT (2020) Genomics of dietary diversification and the evolution of nectar feeding in Phyllostomid bats (Original title: Genómica de la diversificación de las dietas y evolución de la nectarivoría en murciélagos filostómidos). Biomedical Ph.D. Program, Institute of Ecology, Universidad Nacional Autónoma México (UNAM)
Thesis

Gutiérrez-Guerrero YT (2014) The effect of captivity on the genetic diversity of a population of Mexican prairie dog (*Cynomys mexicanus*) [Original title: Efecto del Cautiverio sobre la diversidad genética de una población del perrito de la pradera mexicano (*Cynomys mexicanus*)]. Institute of Ecology, Universidad Nacional Autónoma México (UNAM)
Thesis

Research Experience

2026 **Centro de Investigação em Saúde de Manhiça, CISM, Mozambique**
Teaching. Workshop: "Malaria genomic surveillance, Bioinformatic and Data Analysis"

2023-2024 **El Colegio de la Frontera Sur (ECOSUR), Tabasco, Mexico**
Teaching : "Selection test using genomes"

2023 **University of California, Berkeley**
Teaching. Workshop: "EvoGeno Methods Workshop"

2020-2023 **Faculty of Science, UNAM**
Invited Talk. Course: "Conservation Genetics in Captive Populations"

2020 **Institute of Ecology, UNAM**
Teaching. Course: "Evolutionary Genomics"

2016 **XIII National Mammalogical Congress, Asociacion Mexicana de Mastozoologia A.C., Mexico**
Instructor:"Metagenomics analysis in Mammals"

2015-2014 **Institute of Ecology, UNAM**
Research Asistant. "Project: Analysis of diversity analysis with molecular data of Teocintle species (Zea mays ssp. mexicana and Zea mays ssp. parviglumis)"
Scholarship by the National Council of Science and Technology (CONACyT), Assistant of a National Researcher Level III or Emeritus (grant no. 9592)

Thesis Supervisor

2025 **Supervisor Comittee Member**
ECOSUR, National Council of Science and Technology (CONAHCYT)
Master Honor Thesis
Student. Katia Hernández Bolaños. "Insights into the evolution of hematophagous bats (subfamily Desmodontinae): substitution rates and evolutionary history."

2024 **Co-Supervisor**
European Erasmus Project - MSc MER, Institute of Marine Sciences-CNR, Venice, Italy
Master Thesis Committee
Student. Christine Barrow. "Modelling and prioritisation of climate refugia for marine spatial planning"

2023 **Co-Supervisor**
Faculty of Veterinary and Zootechnics, UNAM
Undergraduate Honor Thesis Committee
Student. Isabela Ruelas Mesa. "In sillico modelling of the coding proteins of Diarrhea Bovine, an evolutive approximation and protein contact prediction of Caspasa-1"

Conferences

Oral Presentations

Gutiérrez-Guerrero YT., et al., (2019) "Genomic shifts behind dietary diversification in Phyllostomid bats and genomic signatures of parallel evolution in nectar-feeders", International Congress of North American Society of Bat Research. (NASBR). Kalamazoo, USA

Gutiérrez-Guerrero YT., et al., (2018) “Whole-genome sequencing reveals signatures of dietary specialization in New World Leaf-Nosed bats”, International Congress of North American Society of Bat Research (NASBR). Puerto Vallarta, Mexico.

Gutiérrez-Guerrero YT., et al., (2017) “How to live on sugar? Genetic mechanisms of the Nectar feeding bat Leptonycteris yerbabuenae”. VI Mexican Congress of Ecology, Leon, Guanajuato, Mexico

Gutiérrez-Guerrero YT., Castellanos-Morales G., Eguiarte-Fruns LE (2014) “Efecto del cautiverio sobre la diversidad genética de una población del perrito de la pradera mexicano (Cynomys mexicanus)”. XII National Mammalogical Congress, Asociación Mexicana de Mastozoología A.C. Puebla, Mexico

Poster Presentation

Gutiérrez-Guerrero YT., Orozco-Arias S., González J., Nachman MW (2024) “Genomic lanscape of transposable elements in wild house mouse populations from different environments”. International Conference of Transposon Elements (ICTE). Saint Malo, France

Gutiérrez-Guerrero YT., Phifer-Rixey M., Martins F, Nachman MW (2023) “Comparing continents: The genomic basis of environmental adaptation and parallel evolution in house mice (Mus musculus domesticus) from the Americas”. Society for Molecular Biology and Evolution (SMBE). Ferrara, Italy

Gutiérrez-Guerrero YT., Phifer-Rixey M., Martins F, Nachman MW (2022) “Genomic basis of climatic adaptation and parallel evolution in house mice from North and South America”. Population, Evolutionary and Quantitative Genetics (PEQG). Asilomar, California, USA

Diversity, Inclusion, Education and Outreach

2023-2021 **Working Group: Scientific Translation (Spanish and Portuguese)**
Museum of Vertebrate Zoology, Department of Integrative Biology, University of California, Berkeley

2023-2021 **Education and Outreach: Museum Tour Guide across vertebrate collections**
Museum of Vertebrate Zoology, Department of Integrative Biology, University of California, Berkeley